

# IOPS Reproducible benchmarking, from one YAML to a report

Integrated Orchestration for Parametric Studies



Describe the study once in YAML. IOPS does the rest:



Sweep: exhaustive, random, **Bayesian**,  
adaptive



Run **locally or on SLURM**, cached, repeated



Parse metrics into an **interactive report**



Archive and re-explore any run

## IO500 Benchmark - Analysis Report

Generated: 2023-05-30 11:32:07  
Benchmark Name: 2023-05-29T11:04:34.112281  
Description:  
Total Executions: 683

### Summary Statistics

#### Execution Overview

| Metric           | Value           |
|------------------|-----------------|
| Benchmark Name   | IO500 Benchmark |
| Search Method    | evaluation      |
| Total Executions | 450 x 3 = 983   |

#### Metrics Overview

| Metric | Min    | Max     | Mean    | Std Dev |
|--------|--------|---------|---------|---------|
| SCORE  | 4.5004 | 15.1851 | 20.3754 | 2.4358  |
| BNW    | 1.7851 | 6.1183  | 3.7170  | 1.0941  |
| MD     | 7.0005 | 45.4533 | 29.4300 | 6.9124  |

#### Execution Details

#### Search Space

### Best Configurations

Top 5 configurations for each metric:

#### Best for SCORE

| Rank                                                     | nodes | processes_per_node | transfer_size_mb | segment_count | files_per_proc | plnd_queue_size | ost_count | file_per_proc | unique_rle | SCORE (avg) | Std Dev | Score |
|----------------------------------------------------------|-------|--------------------|------------------|---------------|----------------|-----------------|-----------|---------------|------------|-------------|---------|-------|
| 1                                                        | 32    | 32                 | 1                | 3000          | 100            | 100             | 8         | True          | True       | 19.1081     | nan     | 1     |
| Command (error rendering command - 'input' is undefined) |       |                    |                  |               |                |                 |           |               |            |             |         |       |
| 2                                                        | 32    | 16                 | 1                | 3000          | 100            | 100             | 8         | True          | False      | 14.9082     | nan     | 1     |

 [iops.gitlabpages.inria.fr](https://iops.gitlabpages.inria.fr) |  `pip install iops-benchmark`